Darren Shen

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EDUCATION

San Jose State University **Graduation: May 2027**
B.S. Computer Engineering, GPA: 3.9/4.0
- **Select Coursework**: Calculus, Discrete Math, C Programming, Data Structures & Algorithms

EXPERIENCE

SJSU College of Engineering **Aug. 2024 – Current**
Undergraduate Research Assistant
- Contributed to open-source project deployed in Ukraine to provide offline regions with internet-like access via Android
- Performed end-to-end testing across Android clients, transport, and server, reducing manual test cycles by 25%
- Integrated message counters into K-9 Mail to track sent/received bundles, increasing offline testing efficiency by 30%
- Fixed a K-9 Mail login bug with a Kotlin-based retry mechanism, improving successful login rates by 40%
- Developed a Kotlin Multiplatform (KMP) app to test iOS and Android communication protocols

San Jose State University **Jun. 2025 – Aug. 2025**
Software Engineering Intern
- Built an RFID card management system for room access using MongoDB, Node.js, Express.js, React, and TailwindCSS, supporting 100+ users
- Stored card bytes with hashing and salting using the bcrypt library, generating aliases for cards for easy reference
- Created a real-time card activity monitor with Server-Sent Events (SSEs), improving debugging efficiency by 30%
- Deployed the website with Docker, minifying the React code and serving static content with Nginx

The Software & Computer Engineering Society **Feb. 2025 – Current**
Software Engineering Intern
- Programmed embedded firmware for GPIO control on a custom six-winged drone using an STM32 microcontroller
- Collaborated with a team of 3 on hardware integration and sensor testing to ensure reliable PCB-based system performance

PROJECTS

GraphGuard | *Next.js, FastAPI, LangGraph, OpenAI*
- Built an AI-powered network security platform that tracks global network traffic and autonomously detects, investigates, and mitigates cyber attacks using multi-agent reasoning
- Created an agent system using LangGraph for detection, real-time monitoring, incident explanation, and mitigation actions
- Produced a streaming pipeline and interactive 3D visualizations in Three.js, mapping network flows as nodes and edges
- Ingested and visualized live NetFlow v5 data as graph models, used CIC-DDoS2019 solely for demo and traffic modeling
- Developed a full-stack application with Next.js, FastAPI, LangGraph, NetFlow v5, and OpenAI

QuakeSafe | *React Native, FastAPI, Amazon S3, Supabase (PostgreSQL), Claude, Groq*
- Deployed an AI-powered computer vision application that analyzes earthquake risk from photos of city infrastructure
- Implemented a Groq-based AI chatbot that simulates an earthquake expert to deliver real-time safety guidance
- Developed an image analysis pipeline with Claude to interpret visuals and generate contextual responses
- Built a cross-platform app using React Native, FastAPI, Amazon S3, Supabase (PostgreSQL)

Syll.ai | *Next.js, FastAPI, Tailwind, Vercel, Supabase (PostgreSQL)*
- Created an AI platform that processes syllabus PDFs, summarizes key course info using a language model, and generates a personalized dashboard with course logistics and grades
- Developed a real-time Python pipeline with GeminiAPI to extract information from unstructured PDFs
- Built a web app with FastAPI, Next.js, Tailwind, Vercel, and Supabase (PostgreSQL)

SKILLS

Languages: Java, Python, C, JavaScript, TypeScript, MATLAB, SQL, Kotlin, HTML/CSS
Frameworks: React.js, Next.js, FastAPI, Tailwind CSS, Android SDK, Node.js, Express.js, LangGraph
Technologies: Supabase (PostgreSQL), Amazon S3, Vercel, GitHub, Docker, MongoDB, Postman